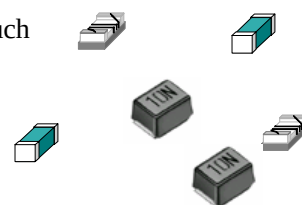


WIRE WOUND CHIP INDUCTORS SPI/SWI/SCI SERIES

Introductions

The SPI/SWI/SCI series are chip inductors widely used in the communication applications such as cellular phones, pagers, and other electronic devices. The wire wound features advance in higher self resonate frequency, better Q factor, and much more stable performance.



Features

- * Operating temperature -40°C to $+85^{\circ}\text{C}$.
- * Excellent solderability and resistance to soldering heat.
- * Suitable for flow and reflow soldering..
- * Good dimensions, high reliability, and easy surface mount assembly.
- * At least 3 types of materials provide wide range of inductance value for flexible needs.

Part Number Code

SPI 0603 C T 3N3 J

1 2 3 TAPING 4 5 Internal Code

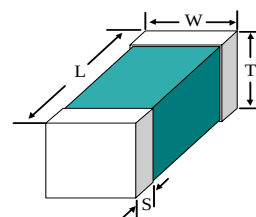
1. Product Type

SPI Series : Laser cut
 SWI Series : Wire Wound
 SCI Series : Molding

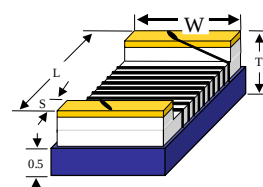
2. Chip Dimension

Size (inch) mm	Length (L) (inch) mm	Width (W) (inch) mm	Thickness (T) (inch) mm	Terminal (S) (inch) mm
SPI 0402 1005	(0.039 ± 0.004) 1.00 ± 0.10	(0.020 ± 0.004) 0.50 ± 0.10	(0.020 ± 0.004) 0.50 ± 0.10	(0.0098 ± 0.004) 0.25 ± 0.10
SPI 0603 1608	(0.063 ± 0.008) 1.60 ± 0.20	(0.031 ± 0.008) 0.80 ± 0.20	(0.031 ± 0.008) 0.80 ± 0.20	(0.016 ± 0.004) 0.30 ± 0.10
SWI 0603 1608	(0.063 ± 0.008) 1.60 ± 0.20	(0.041 ± 0.008) 1.05 ± 0.20	(0.041 ± 0.008) 1.05 ± 0.20	(0.014 ± 0.004) 0.35 ± 0.10
SWI 0805 2012	(0.080 ± 0.008) 2.00 ± 0.20	(0.050 ± 0.008) 1.25 ± 0.20	(0.048 ± 0.008) 1.20 ± 0.20	(0.016 ± 0.004) 0.40 ± 0.10
SWI 1008 2520	(0.098 ± 0.008) 2.50 ± 0.20	(0.063 ± 0.008) 2.00 ± 0.20	(0.063 ± 0.008) 1.60 ± 0.20	(0.020 ± 0.004) 0.50 ± 0.10
SWI 1210 3225	(0.126 ± 0.008) 3.20 ± 0.20	(0.098 ± 0.008) 2.50 ± 0.20	(0.087 ± 0.008) 2.20 ± 0.20	(0.020 ± 0.004) 0.50 ± 0.10
SCI 1210 3225	(0.126 ± 0.008) 3.20 ± 0.20	(0.098 ± 0.008) 2.50 ± 0.20	(0.087 ± 0.008) 2.20 ± 0.20	(0.075 ± 0.004) 1.90 ± 0.10
SCI 1812 4532	(0.180 ± 0.008) 4.50 ± 0.20	(0.126 ± 0.008) 3.20 ± 0.20	(0.126 ± 0.008) 3.20 ± 0.20	(0.102 ± 0.004) 2.60 ± 0.10
SCI 2220 5650	(0.22 ± 0.012) 5.60 ± 0.30	(0.197 ± 0.012) 5.00 ± 0.30	(0.197 ± 0.012) 5.00 ± 0.30	(0.158 ± 0.008) 4.00 ± 0.20

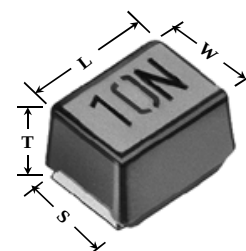
SPI SERIES



SWI SERIES



SCI SERIES



3. Material Type

C : Ceramic

F : Ferrite

H : High Current

4. Inductance Value

3N3 = 3.3 nH

33N = 33 nH

R33 = 330 nH

3R3 = 3.3 uH

330 = 33 uH

331 = 330 uH

5. Tolerance

B = ± 0.15 nH

S = ± 0.3 nH

G = ± 2 %

J = ± 5 %

K = ± 10 %

M = ± 20 %

Specification

Part No.				Inductance ¹ (uH)				Percent Tolerance		Q ² Min		S.R.F. ³ Min (MHZ)		RDC ⁴ Max (OHM)		IDC ⁵ Max (MA)	
SCI	1210 FT	R10	□□□	0.10	@	100	MHZ	J, K, M	30	@	100	MHZ	700		0.44		450
SCI	1210 FT	R12	□□□	0.12	@	25.2	MHZ	J, K, M	30	@	25.2	MHZ	500		0.22		450
SCI	1210 FT	R15	□□□	0.15	@	25.2	MHZ	J, K, M	30	@	25.2	MHZ	450		0.25		450
SCI	1210 FT	R18	□□□	0.18	@	25.2	MHZ	J, K, M	30	@	25.2	MHZ	400		0.28		450
SCI	1210 FT	R22	□□□	0.22	@	25.2	MHZ	J, K, M	30	@	25.2	MHZ	350		0.32		450
SCI	1210 FT	R33	□□□	0.33	@	25.2	MHZ	J, K, M	30	@	25.2	MHZ	300		0.40		450
SCI	1210 FT	R39	□□□	0.39	@	25.2	MHZ	J, K, M	30	@	25.2	MHZ	250		0.45		450
SCI	1210 FT	R47	□□□	0.47	@	25.2	MHZ	J, K, M	30	@	25.2	MHZ	220		0.50		450
SCI	1210 FT	R56	□□□	0.56	@	25.2	MHZ	J, K, M	30	@	25.2	MHZ	180		0.55		450
SCI	1210 FT	R68	□□□	0.68	@	25.2	MHZ	J, K, M	30	@	25.2	MHZ	160		0.60		450
SCI	1210 FT	R82	□□□	0.82	@	25.2	MHZ	J, K, M	30	@	25.2	MHZ	140		0.65		450
SCI	1210 FT	1R0	□□□	1.0	@	7.96	MHZ	J, K, M	30	@	7.96	MHZ	120		0.70		400
SCI	1210 FT	1R2	□□□	1.2	@	7.96	MHZ	J, K, M	30	@	7.96	MHZ	100		0.75		390
SCI	1210 FT	1R5	□□□	1.5	@	7.96	MHZ	J, K, M	30	@	7.96	MHZ	85		0.85		370
SCI	1210 FT	1R8	□□□	1.8	@	7.96	MHZ	J, K, M	30	@	7.96	MHZ	80		0.90		350
SCI	1210 FT	2R2	□□□	2.2	@	7.96	MHZ	J, K, M	30	@	7.96	MHZ	75		1.00		320
SCI	1210 FT	2R7	□□□	2.7	@	7.96	MHZ	J, K, M	30	@	7.96	MHZ	70		1.10		290
SCI	1210 FT	3R3	□□□	3.3	@	7.96	MHZ	J, K, M	30	@	7.96	MHZ	60		1.20		260
SCI	1210 FT	3R9	□□□	3.9	@	7.96	MHZ	J, K, M	30	@	7.96	MHZ	55		1.30		250
SCI	1210 FT	4R7	□□□	4.7	@	7.96	MHZ	J, K, M	30	@	7.96	MHZ	50		1.50		220
SCI	1210 FT	5R6	□□□	5.6	@	7.96	MHZ	J, K, M	30	@	7.96	MHZ	45		1.60		200
SCI	1210 FT	6R8	□□□	6.8	@	7.96	MHZ	J, K, M	30	@	7.96	MHZ	40		1.80		180
SCI	1210 FT	8R2	□□□	8.2	@	7.96	MHZ	J, K, M	30	@	7.96	MHZ	35		2.00		170
SCI	1210 FT	100	□□□	10	@	2.52	MHZ	J, K, M	30	@	2.52	MHZ	30		2.10		150
SCI	1210 FT	120	□□□	12	@	2.52	MHZ	J, K	30	@	2.52	MHZ	20		2.50		140
SCI	1210 FT	150	□□□	15	@	2.52	MHZ	J, K	30	@	2.52	MHZ	20		2.80		130
SCI	1210 FT	180	□□□	18	@	2.52	MHZ	J, K	30	@	2.52	MHZ	20		3.30		120
SCI	1210 FT	220	□□□	22	@	2.52	MHZ	J, K	30	@	2.52	MHZ	20		3.70		110
SCI	1210 FT	270	□□□	27	@	2.52	MHZ	J, K	30	@	2.52	MHZ	20		5.00		80
SCI	1210 FT	330	□□□	33	@	2.52	MHZ	J, K	30	@	2.52	MHZ	17		5.60		70
SCI	1210 FT	390	□□□	39	@	2.52	MHZ	J, K	30	@	2.52	MHZ	16		6.40		65
SCI	1210 FT	470	□□□	47	@	2.52	MHZ	J, K	30	@	2.52	MHZ	15		7		60
SCI	1210 FT	560	□□□	56	@	2.52	MHZ	J, K	30	@	2.52	MHZ	13		8		55
SCI	1210 FT	680	□□□	68	@	2.52	MHZ	J, K	30	@	2.52	MHZ	12		9		50
SCI	1210 FT	820	□□□	82	@	2.52	MHZ	J, K	30	@	2.52	MHZ	11		10		45
SCI	1210 FT	101	□□□	100	@	0.796	MHZ	J, K	20	@	0.796	MHZ	10		11		40

1. Inductance is measured in HP-4285A Precision LCR meter
HP-4287A RF LCR meter with HP-16193 fixture.

2. Q is measured in HP-4285A Precision LCR meter
HP-4287A RF LCR meter with HP-16193 fixture.

3. SRF is measured in HP-8753E RF network analyzer

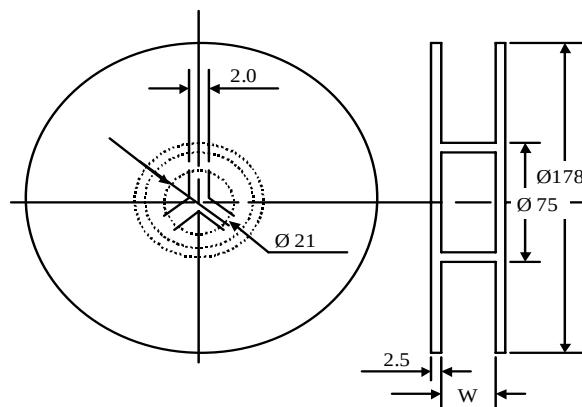
4. RDC is measured in HP-4338B milliohmmeter.

5. For 15 °C Rise.

Packing Quantity

TYPE	PCS/REEL
SPI/SWI 0402	10,000
SPI/SWI 0603	3,000
SWI 0805	2,000
SWI 1008	2,000
SWI 1210	2,000
SCI 1210	1,000
SWI 1812	750
SCI 1812	500

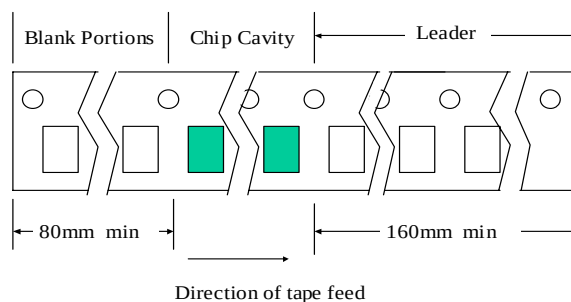
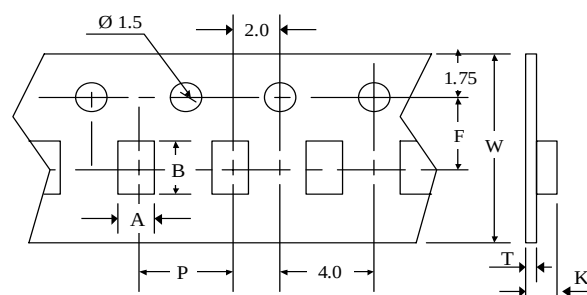
Reel Dimensions



Dimensions (unit: m/m)

TYPE	A	B	P	F	K	T	W
SPI 0402	0.70	1.20	2.00	3.50	0.50	0.70	8.00
SPI 0603	1.00	1.80	4.00	3.50	1.00	0.20	8.00
SWI 0402	0.70	1.20	4.00	3.50	0.50	0.70	8.00
SWI 0603	1.25	1.80	4.00	3.50	1.10	0.20	8.00
SWI 0805	1.42	2.26	4.00	3.50	1.30	0.20	8.00
SWI 1008	2.23	2.73	4.00	3.50	1.80	0.20	8.00
SWI 1210	2.69	3.56	4.00	3.50	2.40	0.20	8.00
SWI 1812	3.60	5.00	8.00	5.50	3.60	0.30	12.00
SCI 1210	2.80	3.60	4.00	3.50	2.40	0.20	8.00
SCI 1812	3.60	5.00	8.00	5.50	3.60	0.30	12.00

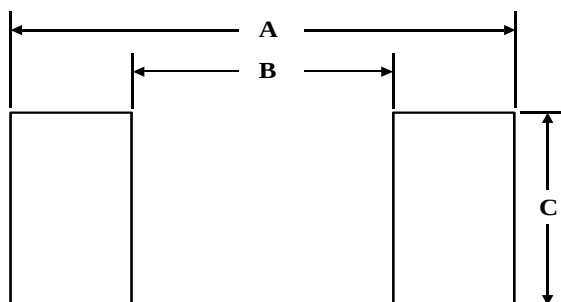
Tape Dimension



Dimensions (unit: m/m)

TYPE	A	B	C
SPI 0402	1.60	0.50	0.05
SPI 0603	2.10	1.00	0.80
SWI 0402	1.20	0.45	0.65
SWI 0603	1.90	0.65	1.00
SWI 0805	2.60	1.20	1.20
SWI 1008	3.80	1.20	2.20
SWI 1210	4.30	2.00	2.82
SWI 1812	5.30	3.00	4.00
SCI 1210	4.00	2.00	2.00
SCI 1210	6.00	3.00	2.80

Recommended Pattern



Remark: 1) Blank length 200 mm minimum for loading
2) Blank length 160 mm minimum for unloading